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Suppressed Immune Response in Infant Monkeys Associated with Maternal Separation[1]

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The effect of maternal separation on an in vitro measure of the cellular immune response was studied in infant bonnet monkeys (*Macaca radiata*). Using a modified mixed-species separation paradigm, bonnet mother–infant pairs were each housed with a single adult female pigtailed monkey (*Macaca nemestrina*). Over a 14-day separation period, the infants showed a suppression of lymphocyte proliferation in response to mitogens relative to a 14-day baseline period which preceded separation. The lymphocyte response was restored following reunion. During separation, infants were initially agitated, and this was followed by observations of reduced activity and a slouched posture. Behaviors returned to baseline conditions following reunion. It was concluded that suppressed immunological functioning may be included among the pathophysiological consequences of maternal separation in infant monkeys.

Separations and losses rank high in life change/stress inventories (Dohrenwend & Dohrenwend, 1974). Grief and bereavement are frequently associated with increased morbidity and mortality among survivors (Fredrick, 1976; Hurst, Jenkins, & Rose, 1976; Sklar & Anisman, 1981). Depression, which often follows loss and separation (Brown, Harris, & Copeland, 1977; Roy, 1981), is associated with a twofold increase in risk of cancer over a 17-year follow-up period (Shekelle, Raynor, Ostfeld, Garron, Bieliavskas, Liv, Maliza, & Paul, 1981). Losses or separations have been suggested as significant antecedents of several childhood disorders such as Graves disease (Morillo & Gardner, 1979), leukemia (Jacobs & Charles, 1980), and rheumatoid arthritis (Henoch, Batson, & Baum, 1978). One hypothesis that might account for the apparent increase in

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vulnerability following a loss or separation is impaired immunological functioning.

Few clinical studies have directly measured immunological functioning associated with grief and bereavement. In a retrospective study, the ability of lymphocytes to undergo blast formation in response to mitogens, an indicator of cellular immune functioning, was suppressed in survivors 6 weeks following the death of the spouse (Bartrop, Lazarus, Luckhurst, Kiloh, & Penny, 1977). In a prospective report, it was observed that the lymphocyte response was suppressed as early as 2 weeks following the death of a spouse (Schleifer, Keller, McKegney, & Stein, 1980). We have recently reported similar prospective observations following peer separation of young pigtailed (*Macaca nemestrina*) monkeys (Reite, Harbeck, & Hoffman, 1981). These studies strongly support the notion that cellular immune function is altered following a loss or separation.

The maternal separation paradigm in infant monkeys represents a valuable animal model system for studying the consequences of loss and separation (Kaufman, 1978; McKinney, 1977; Mineka & Suomi, 1978; Reite, 1977). Since the nature of surrogate and peer attachment bonds in monkeys may be different from the mother–infant attachment bond (Hennessey, Kaplan, Mendoza, Lowe, & Levine, 1979; Reite, Short, & Seiller, 1978), we wanted to extend our previous observations of peer-separated infants to the effects of disruption of the mother–infant bond.

The present pilot study assessed in vitro lymphocyte proliferative responses during maternal separation in both mother and infant bonnet (*Macaca radiata*) monkeys using a modification of previous separation paradigms. Bonnet infants exhibit a behavioral and physiological response to maternal separation similar to pigtailed infants if the separation is conducted in a mixed-species group in which all conspecifics and mother are removed (Kaufman & Stynes, 1978; Reite & Snyder, 1982). Thus for the present study, we established a "mini" mixed-species group in which a bonnet mother–infant pair was housed with a single adult, female pigtailed monkey. During separation, the infant was housed with only the pigtail monkey in order to facilitate obtaining blood samples.

METHODS

Two bonnet monkey mother–infant pairs were tested. Both infants were the first born for each mother, both of which were also lab born. A male infant (2.8.1), 30 weeks old when testing began, and its mother (2.8), 8 years old, were housed with a wild-born multiparous pigtailed monkey (19), whose age was estimated to be 12–13 years. A female infant (6.2.5.1), 35 weeks old when testing began, and its mother (6.2.5), 7 years old, were housed with a 9-year-old, lab-born, nulliparous pigtailed monkey (8.2). The test cage (2.3 × 0.7 × 0.8 m), which housed both the bonnet monkey mother–infant pair and the pigtailed monkey, was

divided equally by a Plexiglas partition with a small opening at the bottom. The opening in the Plexiglas partition was large enough to allow either the bonnet mother or infant to move freely between sides. However, the adult pigtailed monkey was restricted to one side due to her larger size, thus permitting the bonnet monkeys to escape any aggressive encounters with the larger pigtailed monkey. However, aggression was rare for both pigtailed monkeys. The cage also provided a squeeze capacity for rapid capture of the animals for blood sampling. Water was available ad libitum. Feeding (Wayne Monkey Chow and fresh fruit) took place at 0930 hr. Lights were on from 0700 to 1900 hr.

The bonnet mothers and infants were bled via a superficial arm or leg vein between 0800 and 0900 hr on Monday and Friday of each week throughout the entire experiment. Heparinized blood samples (1.5–2.0 ml) were centrifuged for 20 min at 400*g*, and the plasma was removed. The plasma-depleted whole blood was reconstituted to its original volume in sterile Dulbecco's balanced salt solution.

The lymphocytes were separated under sterile conditions by Ficoll–Hypaque sedimentation (Boyum, 1968) and adjusted to a final concentration of 1×10^6 mononuclear cells/ml of RPMI-1640 medium (supplemented with 100 U/ml penicillin, 100 μg/ml streptomycin, 2 mM L-glutamine, and 5% heat-inactivated fetal calf serum). The number of mononuclear cells recovered ranged from 1.8×10^6 to 8.2×10^6. When cell recovery was poor, only the optimum concentration of the mitogen was used (see Table 1) in order to assure triplicate cultures. Lymphocyte stimulation was performed on the day the blood was obtained; blood samples were not pooled. The mitogens concanavalin A (ConA), phytohemagglutinin (PHA), and pokeweed (PWM) were used to stimulate blast formation by the lymphocytes. Basically, 0.1 ml of 1×10^6 cells/ml of supplemented RPMI-1640 was added in triplicate to wells of a microtiter plate for each concentration of the mitogens tested and control cultures to which no

TABLE 1
Average [^{3}H]Thymidine Incorporation [(Experimental-Control Counts) $\times 10^{-3}$] as a Function of Mitogen Concentration

Mitogen added (μg/well)	Mitogen tested		
	PHA	ConA	PWM
0.01	—	13.81 ± 8.56[a]	—
0.02	14.93 ± 9.89	—	—
0.10	38.38 ± 18.71	33.41 ± 14.97	4.14 ± 3.47
0.50	46.63 ± 24.77	82.09 ± 43.97	—
1.00	18.49 ± 9.72	61.18 ± 31.80	13.36 ± 8.58
2.00	—	—	13.08 ± 8.51

[a] Mean ± SD, $N = 16$.

mitogen was added. Cultures were incubated for 3 days at 37°C in a humidified 5% CO_2 incubator. Cultures were then pulsed with 1 μCi/well of [*methyl*-^{3}H]thymidine (New England Nuclear), and 4 hr later, cultures were harvested onto glass-fiber filter paper. The incorporation of [^{3}H] thymidine into newly synthesized DNA was determined by counting cells obtained from stimulated and unstimulated cultures in a liquid scintillation counter. Results are expressed as a stimulation index for the optimum mitogen concentration. The stimulation index (*E/C*) is found by dividing the average cpm of triplicate measures for the mitogen-stimulated cultures by the average cpm of triplicate measures for the unstimulated control cultures.

Several concentrations of each mitogen were tested in 16 monkeys to determine an optimum mitogen concentration for production of maximum lymphocyte proliferation. Table 1 gives the average [^{3}H]thymidine incorporated based on observations of 7 adult female and 9 infant (5 male and 4 female) monkeys for varied concentrations of the mitogens tested. Four of these monkeys were used as subjects in the present separation study. No differences were noted between adults and infants of the age range (4–7 months) tested.

The mother–infant pairs to be tested were removed from their social group to the experimental cages. Ten days were allowed for adaptation to the new housing before any measurements were begun. There were no obvious behavioral effects of removing these animals from their social group other than limited activity and reduced social interactions. At this time, quantified behavioral observations (Reite & Short, 1980) were begun and made twice daily (5 min/monkey/observation) on weekdays only, until the completion of the study, using an on-line real-time LSI-11/23-based computer system, which has been described in detail elsewhere (Walker & Reite, 1978). A computer malfunction destroyed baseline and early separation observations; as a result we can present qualitative observations of behavioral change only.

Both immunological and behavioral data were obtained during a 2-week baseline period (following the 10-day adaptation period), at which time the mothers were removed to another room out of sight, sound, and smell of the infant. The mothers were housed in individual cages in a room containing other individually housed animals. The infants remained alone with the pigtailed adult females. The separation period lasted 2 weeks at the end of which time the mothers were returned to the infants. Behavioral and immunological measures were obtained for an additional 2 weeks after the reunion. During separation no behavioral observations were made on the mothers, however, blood samples were still obtained.

RESULTS AND DISCUSSION

Maternal separation was initially associated with agitated behavior on the part of both infants based on qualitative observations. This included

increased activity and vocalizations. Activity declined during the later part of separation. The activity of one infant (6.2.5.1) was particularly low, and this was frequently associated with slouched postures. A slouched posture was noted in both infants throughout the separation period. The assumption of slouched postures is an important indicator of behavioral depression in separated infant monkeys (Reite, Short, Seiler, & Pauley, 1981). Attempts by the infants to initiate interactions with the pigtailed females through eye contact and physical contact were common during the separation period. However, these attempts were ignored by both pigtailed females. At reunion, both infants returned immediately to their mothers. Slouched postures were no longer observed following reunion.

Maternal separation was associated with depressed lymphocyte proliferative responses to the optimal concentrations of PHA (0.5 μg/well) and ConA (0.5 μg/well) in both infants. Little change in the stimulation index was noted to the optimal concentration of PWM (1.0 μg/well). Stimulation indices returned to baseline values following reunion for both infants. One mother (2.8) showed a reduced stimulation index during separation which returned to baseline values following reunion; however, the other mother (6.2.5) showed a declining trend in lymphocyte response which continued into the reunion period. These results are shown in Figs. 1 and 2.

Our results replicate the previous finding of disturbed immune functioning associated with peer separation in pigtailed infants (Reite et al., 1981), and they demonstrate as well that the effect is not species specific to

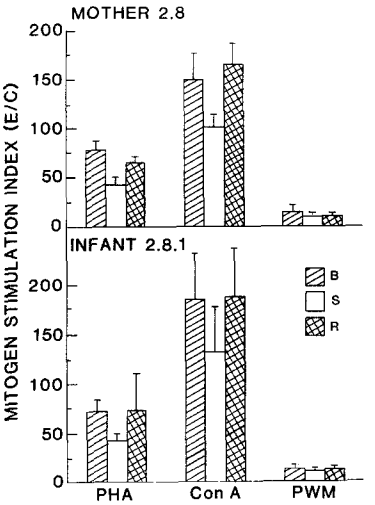


FIG. 1. Average mitogen stimulation indices for optimum mitogen concentrations based on Table 1 for monkeys 2.8 (mother) and 2.8.1 (infant) during baseline (B), separation (S), and reunion (R). Average is determined from results of four blood samples obtained on four different days for each condition. Brackets indicate one SEM.

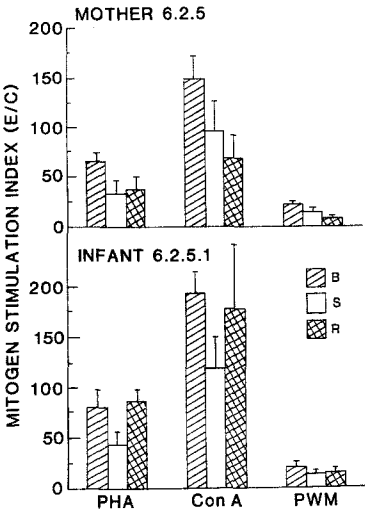


FIG. 2. Average mitogen stimulation indices for monkeys 6.2.5 (mother) and 6.2.5.1 (infant). See Fig. 1 for details.

M. nemestrina. The present results suggest that maternal separation in bonnet infants impairs an in vitro lymphocyte response to the T-cell (thymus-derived lymphocyte) mitogens PHA and ConA. It is generally thought that PWM primarily stimulates B cells (bone marrow-derived lymphocytes) in vitro (Daguillar, 1972). Perhaps the ability of B cells to proliferate in vitro was not affected by the separation experience. However in mice, PWM may also stimulate T cells (Janossy & Greaves, 1971), and one cannot conclude that only B cells were unaffected. These issues must be resolved in future studies. These pilot data also suggest that an alteration in immune functioning of the mother may occur during separation from her infant. However, one must be cautious in interpreting the adult data since the change in housing and not the separation itself may have in part been responsible for the suppression (Hill, Greer, & Felsenfeld, 1967; Riley, 1981). These questions can be answered by removing the infant rather than the mother from the group and observing immunological functioning of the group-housed, separated mother.

The nature of the link between impaired lymphocyte proliferation and maternal separation is unclear. Stress, in a variety of forms, is frequently associated with adrenocortical activation (Riley, 1981), and the immunosuppressive effects of corticosteroids as well documented (Claman, 1975). Prolonged adrenocortical activation does not seem to be a predominant response to maternal separation in infant rhesus monkeys (Gunnar, Gonzalez, Goodlin, & Levine, 1981). In the present study, maximum suppression of in vitro lymphocyte proliferation was noted 8–12 days following separation, at a time when cortisol levels were no different from baseline

(Gunnar et al., 1981). The duration of the immunosuppressive effects of the high cortisol levels noted early in separation are unknown. However, it is not certain that the adrenal response to maternal separation in bonnet monkeys is the same as in rhesus monkeys and generalizations to the present study may not be appropriate.

Alterations in immune functioning associated with maternal separation could be produced by a different or additional mechanism. Several reports have demonstrated through a variety of techniques evidence for sympathetic innervation of lymphoid tissues (Bulloch & Moore, 1981; Felten, Overhage, Felten, & Schmedtje, 1981; Giron, Crutcher, & Davis, 1980; Williams & Felten, 1981; Williams, Peterson, Shea, Schmedtje, Bauer, & Felten, 1981). It was suggested that altered catecholamine levels in lymphoid tissues, modulated in addition by catecholamine release from the adrenals, might act directly on lymphocytes thereby altering immunoresponsiveness (Williams et al., 1981). Interestingly, enzymes involved in catecholamine biosynthesis (tyrosine hydroxylase, dopamine-β-hydroxylase, and phenylethanolamine-*N*-methyltransferase) increase in infant rhesus monkeys during the agitation phase following maternal separation (Breese, Smith, Mueller, Howard, Prange, Lipton, Young, McKinney, & Lewis, 1973). The extent to which catecholamines may have affected our observations remains to be determined.

These data suggest that disturbances in cellular immune function are one component of the complex series of physiological changes accompanying loss or separation (Reite et al., 1981). They support the hypothesis that the increase in vulnerability following loss or separation may be due in part to alterations in immunological functioning. We recognize that the small number of subjects tested to date must be increased to further document our hypothesis, and additional measures of immune functioning must be included in future studies.

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